

Regularisation Theory

What is a regularisation and why do we regularise?

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- Regularisation Theory – What is a regularisation and why do we regularise?

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- Learning optimal sampling strategies for Magnetic Resonance Imaging (MRI)

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- Lifted training and inversion of neural networks

Focus of First Lecture(s)

Regularisation theory

What is a regularisation and why do we regularise?

Contents

- 1 Introduction to Inverse Problems
- 2 Fundamental Concepts in Regularisation
- 3 Selecting Solutions
- 4 Convergent Regularisation Methods
- 5 Variational Regularisation Methods
- 6 Convergence Analysis: Error Estimates
- 7 Iterative Regularisation Methods
- 8 Data-Driven Regularisation: Spectral Methods
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Introduction to Inverse Problems

Introduction: What is an Inverse Problem?

General Form

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for u , where

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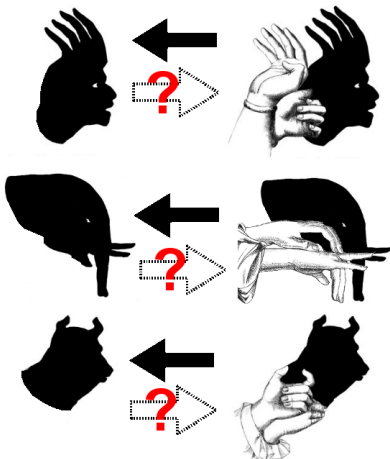
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Some useful references: [5, 14, 1]

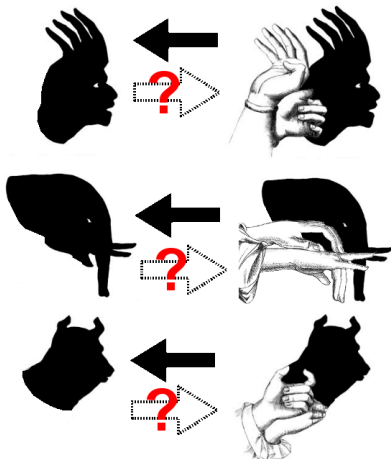
Inverse Problems: A Visual Illustration



The Shadow Image Problem

- **Forward Problem:** Given 3D hand shapes, compute 2D shadows

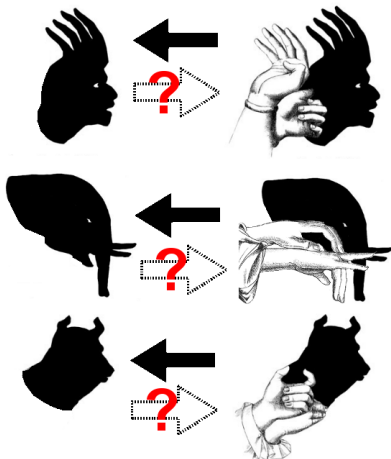
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Inverse Problems: A Visual Illustration



The Shadow Image Problem

- **Forward Problem:** Given 3D hand shapes, compute 2D shadows
- **Inverse Problem:** Given only the 2D shadow silhouettes, determine the original 3D object
- Multiple different 3D objects can produce same shadow (non-uniqueness)

The Challenge: Well-Posedness vs. Ill-Posedness

Conditions for Well-Posedness

Most practical inverse problems are *ill-posed* in the sense of Hadamard [7, 8] and John [9]. A problem is *well-posed* if it satisfies

- 1 **Existence:** A solution u exists for all $f \in \mathcal{V}$.
- 2 **Uniqueness:** The solution is unique.
- 3 **Stability (Continuity):** The solution u depends continuously on the data f . Small changes in f lead to small changes in u .

If any of these conditions are violated, the problem is *ill-posed*. Stability is often the most problematic condition in practice.

Example: The Inverse Problem of Differentiation

Problem Setup

Consider the task of finding a function $u(x)$ given its integral $f(y)$. If we assume $f(0) = 0$, we want to find $u = f'$. This can be formulated as solving $Ku = f$ where the operator K is integration, i.e.

$$(Ku)(y) = \int_0^y u(x) \, dx = f(y).$$

Here, $K : C([0, 1]) \rightarrow \{g \in C^1([0, 1]) \mid g(0) = 0\}$ (or suitable L^p spaces, e.g., $L^2([0, 1]) \rightarrow L^2([0, 1])$).

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Goal

Our goal is to recover u (the derivative) from f (the integral). This is the inverse problem of differentiation.

III-Posedness of Differentiation: Setup

Suppose instead of the exact data f , we observe noisy data $f^\delta = f + n^\delta$, where n^δ is some noise term. We are interested in $u^\delta = (f^\delta)' = f' + (n^\delta)' = u + (n^\delta)'$.

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A Perturbation Example

Consider the sequence of noise functions $n^\delta \in L^\infty([0, 1])$:

$$n^\delta(x) := \delta \sin\left(\frac{kx}{\delta}\right)$$

for a fixed but arbitrary number $k > 0$.

- The noise in the data can be made arbitrarily small:

$$\|n^\delta\|_{L^\infty([0,1])} = \delta \rightarrow 0 \quad \text{as } \delta \rightarrow 0$$

III-Posedness of Differentiation: Analysis

A Perturbation Example (continued)

- However, the derivative of this noise is:

$$(n^\delta)'(x) = k \cos\left(\frac{kx}{\delta}\right)$$

- The error in the reconstructed derivative u^δ is:

$$\|u - u^\delta\|_{L^\infty([0,1])} = \|(n^\delta)'\|_{L^\infty([0,1])} = k$$

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Conclusion on Ill-Posedness

Despite noise becoming arbitrarily small ($\|n^\delta\|_{L^\infty} \rightarrow 0$), the error in u^δ remains k . Therefore, u^δ does not depend continuously on f^δ in the L^∞ norm. The problem is ill-posed.

The Operator's "Blueprint": Singular Value Decomposition (SVD)

Many inverse problems involve *compact linear operators* $K : \mathcal{U} \rightarrow \mathcal{V}$ between Hilbert spaces. The SVD provides a fundamental way to understand the action of such operators.

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Singular Value Decomposition (SVD) of K

For a compact operator K , there exist:

- **Singular values:** A sequence $\sigma_1 \geq \sigma_2 \geq \dots > 0$. These are positive numbers that typically decay towards zero ($\sigma_j \rightarrow 0$) if the range of K is infinite-dimensional.

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- **Singular values:** A sequence $\sigma_1 \geq \sigma_2 \geq \dots > 0$. These are positive numbers that typically decay towards zero ($\sigma_j \rightarrow 0$) if the range of K is infinite-dimensional.
- **Orthonormal sets** (or bases for relevant subspaces):
 - $\{u_j\}_{j \in \mathbb{N}}$ in $\mathcal{U}$ (input space elements, form an orthonormal basis of $\mathcal{N}(K)^\perp$).
 - $\{v_j\}_{j \in \mathbb{N}}$ in $\mathcal{V}$ (output space elements, form an orthonormal basis of $\overline{\mathcal{R}(K)}$).

SVD: How K Acts

The SVD components are linked by these key relationships:

- $Ku_j = \sigma_j v_j$ (The operator K maps u_j to v_j , scaled/attenuated by σ_j).
- $K^*v_j = \sigma_j u_j$ (The adjoint operator K^* maps v_j back to u_j , also scaled by σ_j).

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Representation of K's Action

Any $w \in \mathcal{U}$ can be represented using the u_j , and K acts on w as

$$Kw = \sum_{j=1}^{\infty} \sigma_j \underbrace{\langle w, u_j \rangle_{\mathcal{U}}}_{\text{coeff. of } w \text{ along } u_j} v_j$$

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SVD and Ill-Posedness

Decay of singular values $\sigma_j \rightarrow 0$ is primary source of ill-posedness for compact operators. To find u from $Ku = f$, one might think of $u \approx \sum \frac{1}{\sigma_j} \langle f, v_j \rangle u_j$. If σ_j become very small, any noise in $\langle f, v_j \rangle$ gets greatly amplified.

SVD for Integration: The Forward Operator

We revisit the inverse problem of differentiation. The forward operator $K : L^2([0, 1]) \rightarrow L^2([0, 1])$ is the integration operator

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This can be written as an integral operator with kernel $k(x, y)$, i.e.

$$(Ku)(y) = \int_0^1 k(x, y) u(x) dx, \quad \text{where } k(x, y) = \begin{cases} 1 & \text{if } x \leq y \\ 0 & \text{if } x > y \end{cases} .$$

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This operator K is compact. Our goal is to find its SVD.

Computing the Adjoint Operator K^*

The adjoint operator K^* is defined by $\langle Ku, v \rangle_{L^2([0,1])} = \langle u, K^*v \rangle_{L^2([0,1])}$. We observe

$$\begin{aligned} \langle Ku, v \rangle &= \int_0^1 \left(\int_0^y u(x) dx \right) v(y) dy \\ &= \int_0^1 \int_0^1 k(x, y) u(x) v(y) dx dy \\ &= \int_0^1 u(x) \left(\int_x^1 v(y) dy \right) dx \end{aligned}$$

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Thus, the adjoint operator $K^* : L^2([0, 1]) \rightarrow L^2([0, 1])$ is given by

$$(K^*v)(x) = \int_x^1 v(y) dy .$$

The Operator K^*K and Eigenvalue Problem

Next, we form the operator K^*K , i.e.

$$\begin{aligned} (K^*Ku)(x) &= K^* ((Ku)(\cdot)) (x) \\ &= \int_x^1 \left(\int_0^y u(z) dz \right) dy . \end{aligned}$$

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We seek eigenvalues $\lambda = \sigma^2 > 0$ and eigenfunctions $u \in L^2([0, 1])$ for K^*K , i.e.

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This leads to the solving the integral equation

$$\int_x^1 \left(\int_0^y u(z) dz \right) dy = \lambda u(x) .$$

Deriving the ODE and Boundary Conditions

- Differentiating with respect to x (using Leibniz integral rule) yields

$$\lambda u'(x) = - \int_0^x u(z) dz .$$

From this, setting $x = 0$, we get $u'(0) = 0$.

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Summary of ODE Problem

We need to solve $\lambda u''(x) + u(x) = 0$ with boundary conditions $u'(0) = 0$ and $u(1) = 0$.

Singular Values σ_j and Orthonormal Functions u_j

The general solution to $u''(x) + \frac{1}{\lambda}u(x) = 0$ is $u(x) = c_1 \sin(x/\sqrt{\lambda}) + c_2 \cos(x/\sqrt{\lambda})$. Let $\sigma := \sqrt{\lambda}$.

- Apply $u'(0) = 0$: $u'(x) = \frac{c_1}{\sigma} \cos(x/\sigma) - \frac{c_2}{\sigma} \sin(x/\sigma)$. $u'(0) = \frac{c_1}{\sigma} = 0 \implies c_1 = 0$. So, $u(x) = c_2 \cos(x/\sigma)$.

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- Apply $u(1) = 0$: $c_2 \cos(1/\sigma) = 0$. For non-trivial solutions ($c_2 \neq 0$), we require $\cos(1/\sigma) = 0$. This means $1/\sigma = (j - \frac{1}{2})\pi$ for $j \in \mathbb{N}$.

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The singular values are $\sigma_j = \frac{1}{(j - \frac{1}{2})\pi} = \frac{2}{(2j - 1)\pi}$ for $j \in \mathbb{N}$. The corresponding normalised eigenfunctions $u_j(x)$ (after choosing $c_2 = \sqrt{2}$ for normalisation) are

$$u_j(x) = \sqrt{2} \cos \left(\left(j - \frac{1}{2} \right) \pi x \right).$$

Computing Orthonormal Functions v_j

The singular functions v_j are obtained via $v_j = \frac{1}{\sigma_j} Ku_j$, i.e.

$$\begin{aligned} (Ku_j)(x) &= \int_0^x u_j(y) dy = \int_0^x \sqrt{2} \cos \left(\left(j - \frac{1}{2} \right) \pi y \right) dy, \\ &= \sqrt{2} \left[\frac{\sin \left(\left(j - \frac{1}{2} \right) \pi y \right)}{\left(j - \frac{1}{2} \right) \pi} \right]_0^x = \sqrt{2} \frac{\sin \left(\left(j - \frac{1}{2} \right) \pi x \right)}{\left(j - \frac{1}{2} \right) \pi}, \\ &= \sigma_j \sqrt{2} \sin \left(\left(j - \frac{1}{2} \right) \pi x \right). \end{aligned}$$

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Therefore,

$$v_j(x) = \frac{1}{\sigma_j} (Ku_j)(x) = \sqrt{2} \sin \left(\left(j - \frac{1}{2} \right) \pi x \right).$$

SVD Summary for Integration Operator

For the integration operator $K : L^2([0, 1]) \rightarrow L^2([0, 1])$, $(Ku)(y) = \int_0^y u(x) dx$:

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Expansion of Kw

$$\begin{aligned} (Kw)(x) &= \sum_{j=1}^{\infty} \frac{2}{(2j-1)\pi} \left(\int_0^1 w(s) \sqrt{2} \cos\left(\left(j - \frac{1}{2}\right)\pi s\right) ds \right) \sqrt{2} \sin\left(\left(j - \frac{1}{2}\right)\pi x\right) \\ &= \sum_{j=1}^{\infty} \frac{4}{(2j-1)\pi} \left(\int_0^1 w(s) \cos\left(\left(j - \frac{1}{2}\right)\pi s\right) ds \right) \sin\left(\left(j - \frac{1}{2}\right)\pi x\right) \end{aligned}$$

Fundamental Concepts in Regularisation

Motivation for Regularisation

Why Regularisation?

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- Since many inverse problems $Ku = f$ are ill-posed (especially regarding stability), direct inversion or naive solutions are often highly sensitive to noise in the data f .
- Regularisation methods aim to find stable approximate solutions by incorporating prior knowledge or preferences about the solution u .
- We need a formal framework to define what constitutes a "good" regularisation.

General Regularisation Operators: Definition

Let $\mathcal{U}, \mathcal{V}$ be metric spaces. For $x \in X, S \subset X$, define $d(x, S) := \inf_{v \in S} d(x, v)$.

Regularisation Operator [1]

Set-valued operators $R_\alpha : \mathcal{V} \rightrightarrows \mathcal{U}$ (parameterised by $\alpha \in A \subset \mathbb{R}^m$) are called **regularisation operators** if for each fixed $\alpha \in A$ and for all $f^\delta \in \mathcal{V}$ and sequences $f^{\delta_n} \in \mathcal{V}$ converging to f^δ (i.e., $d_{\mathcal{V}}(f^{\delta_n}, f^\delta) \rightarrow 0$), we have

$$\emptyset \neq \left\{ u \in \mathcal{U} \mid \limsup_{k \rightarrow \infty} d_{\mathcal{U}}(u, R_\alpha(f^{\delta_k})) = 0 \right\} \subset R_\alpha(f^\delta)$$

The set in the middle is the Kuratowski limit inferior of the sequence of sets $R_\alpha(f^{\delta_n})$.

General Regularisation Operators: Definition

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Regularisation Method

A regularisation operator R_α together with a **parameter choice strategy** $\alpha_{\text{choice}} : (0, \delta_0) \times \mathcal{V} \rightarrow A$, denoted $\alpha(\delta, f^\delta)$, forms a **regularisation method**.

Regularisation Operators: Single-Valued Case

Simplification for Single-Valued Operators

If $R_\alpha : \mathcal{V} \rightarrow \mathcal{U}$ is a *single-valued* operator for each $\alpha \in \Lambda$ then for R_α to be a regularisation operator the previous condition simplifies to R_α being **continuous** on $\mathcal{V}$.

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Verification: If R_α is continuous and $u_n = R_\alpha(f^{\delta_n}) \rightarrow R_\alpha(f^\delta) = u$, then $\{v \mid \lim_k \|v - u_k\|_{\mathcal{U}} = 0\} = \{u\}$. The stability condition becomes $\emptyset \neq \{u\} \subset \{u\}$, which is true.

Example: Spectral Regularisation Operators

We will now examine a common family of regularisation operators known as **spectral regularisation operators**.

General Form of Spectral Regularisation

For a compact linear operator $K : \mathcal{U} \rightarrow \mathcal{V}$ between Hilbert spaces with SVD $\{(\sigma_j, u_j, v_j)\}$, spectral regularisation operators $R_\alpha : \mathcal{V} \rightarrow \mathcal{U}$ are defined by

$$R_\alpha f = \sum_{j=1}^{\infty} g_\alpha(\sigma_j) \langle f, v_j \rangle_{\mathcal{V}} u_j$$

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- $\alpha > 0$ is the regularisation parameter.
- $g_\alpha : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{\geq 0}$ are known as *filter functions*. They modify the way singular values contribute to the reconstruction.
- These operators R_α are linear and single-valued.

Spectral R_α as Regularisation Operators

Recall from the previous slide: for a single-valued operator $R_\alpha : \mathcal{V} \rightarrow \mathcal{U}$ to be a **regularisation operator**, it must be **continuous** on $\mathcal{V}$ for each fixed α .

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Condition for Continuity of Spectral R_α

A spectral regularisation operator R_α is continuous if its associated filter function $g_\alpha(\sigma)$ is bounded for fixed α , i.e.

$$\sup_{\sigma > 0} |g_\alpha(\sigma)| \leq C_\alpha < \infty.$$

If this holds, R_α is a bounded linear operator, because

$$\|R_\alpha f\|_{\mathcal{U}}^2 = \sum_{j=1}^{\infty} |g_\alpha(\sigma_j)|^2 |\langle f, v_j \rangle_{\mathcal{V}}|^2 \leq C_\alpha^2 \sum_{j=1}^{\infty} |\langle f, v_j \rangle_{\mathcal{V}}|^2 \leq C_\alpha^2 \|f\|_{\mathcal{V}}^2.$$

Thus, $\|R_\alpha\|_{\mathcal{L}(\mathcal{V}, \mathcal{U})} \leq C_\alpha$, which implies that R_α is continuous.

Spectral Regularisation Methods

- If a spectral operator R_α is continuous for a fixed α (due to its filter function $g_\alpha(\sigma)$ being bounded by C_α), it qualifies as a **regularisation operator**.

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Example: Tikhonov Regularisation [17, 16, 15]

The filter function for Tikhonov regularisation is $g_\alpha(\sigma) = \sigma/(\sigma^2 + \alpha)$.

- For any fixed $\alpha > 0$, this function is bounded:

$$\sup_{\sigma > 0} \left| \frac{\sigma}{\sigma^2 + \alpha} \right| = \frac{1}{2\sqrt{\alpha}} =: C_\alpha < \infty$$

- Hence, for each $\alpha > 0$, the operator R_α is a continuous regularisation operator.
- Paired with $\alpha(\delta, f^\delta)$, $(R_\alpha, \alpha(\delta, f^\delta))$ forms the *Tikhonov regularisation method*.

Tikhonov Regularisation in Spectral Form

Tikhonov Filter Function

Applied to our integration operator $(Ku)(y) = \int_0^y u(x) dx$, we have

$$(R_\alpha f)(x) = \sum_{j=1}^{\infty} \frac{4(2j-1)^2\pi^2}{4 + \alpha(2j-1)^2\pi^2} \left(\int_0^1 f(s) \sin \left(\left(j - \frac{1}{2} \right) \pi x \right) ds \right) \cos \left(\left(j - \frac{1}{2} \right) \pi x \right) .$$

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General Tikhonov Form

Note: for any bounded linear operator $K : \mathcal{U} \rightarrow \mathcal{V}$ between Hilbert spaces, Tikhonov regularisation can be written as

$$R_\alpha(f^\delta) = (K^*K + \alpha I)^{-1} K^* f^\delta$$

This form doesn't require knowledge of the SVD and applies generally.

Tikhonov as an Optimisation Problem

Variational Formulation

The Tikhonov regularised solution $R_\alpha(f^\delta)$ is also the unique minimiser of

$$R_\alpha(f^\delta) = \arg \min_{u \in \mathcal{U}} \left\{ \frac{1}{2} \|Ku - f^\delta\|_V^2 + \frac{\alpha}{2} \|u\|_U^2 \right\} .$$

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- **Data fidelity term:** $\frac{1}{2} \|Ku - f^\delta\|_V^2$ ensures the solution fits the observed data.

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- **Regularisation term:** $\frac{\alpha}{2} \|u\|_{\mathcal{U}}^2$ penalises large solution norms.
- **Balance:** Parameter $\alpha > 0$ controls trade-off between data fidelity and regularity.
- This variational approach is equivalent to both expression $(K^*K + \alpha I)^{-1} K^* f^\delta$ and spectral form shown on previous slide.

Example: Neural Network

Pre-trained Neural Network

Consider a pre-trained feed-forward neural network $\mathcal{N}_\theta : \mathcal{V} \rightarrow \mathcal{U}$, i.e.

$$R_\theta(f^\delta) := \mathcal{N}_\theta(f^\delta) = W_L \sigma_{L-1}(\dots W_2 \sigma_1(W_1 f^\delta + b_1) + b_2 \dots) + b_L$$

The parameters $\theta = \{(W_j, b_j)_{j=1}^L\}$ are fixed. R_θ is single-valued.

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The parameter α in the definition corresponds to the fixed set of weights θ .

- Each layer (affine transformation $W_j(\cdot) + b_j$ and activation σ_j) is typically continuous. Standard activation functions (ReLU, sigmoid, tanh, etc.) are continuous.

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- Therefore, $R_\theta(f^\delta) = \mathcal{N}_\theta(f^\delta)$ (for fixed pre-trained θ) is a **regularisation operator**.

Beyond Stability: The Need for Convergence

We've established that *regularisation operators* R_α provide stable processing of data for a *fixed* parameter α . A *regularisation method* $(R_\alpha, \alpha(\delta, f^\delta))$ then uses a rule to select α .

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The Next Step: Defining the Target

Before we can formally discuss *convergence* of a regularisation method to the true solution, we need to define what this *true solution* is. This leads us to consider concepts like *best approximate solutions* and *selection operators*.

Selecting Solutions

Best Approximate Solutions and Selection Operators

Best Approximate Solution

Given an error measure $F : \mathcal{V} \times \mathcal{V} \rightarrow \mathbb{R}_+ \cup \{+\infty\}$, we call $\hat{u} \in \mathcal{U}$ a **best approximate solution** of $Ku = f$ with respect to F if

$$F(K\hat{u}, f) \leq F(Ku, f) \quad \text{for all } u \in \mathcal{U}$$

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Selection Operator

A multivalued operator $\mathcal{S} : \mathcal{R}(K) \rightrightarrows \mathcal{U}$ is called a **selection operator** if $\mathcal{S}(Ku) \subset \{u\} + \mathcal{N}(K)$ for all $u \in \mathcal{U}$.

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- Often, $\mathcal{S}(f')$ selects solutions from the set of best approximate solutions for data f' by minimising a secondary (regularisation) functional.

Examples of Selection Operators

Let $K : L^2(\Omega) \rightarrow L^2(\Sigma)$ or similar Hilbert spaces.

Selection via Exact Fit (Minimum Norm)

$$\text{If } F(Ku, f) = \chi_{=0}(f - Ku) = \begin{cases} 0 & Ku = f \\ \infty & \text{else} \end{cases}.$$

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$$S(f) = \arg \min_{u \in L^2(\Omega)} \left\{ \|u\|_{L^2(\Omega)} \mid Ku = f \right\}$$

This yields

$$S(f) = \begin{cases} \{K^\dagger f\} & \text{if } f \in \mathcal{R}(K) \\ \emptyset & \text{if } f \notin \mathcal{R}(K) \end{cases},$$

where $K^\dagger$ is the Moore-Penrose pseudo-inverse of K .

Examples of Selection Operators (continued)

Selection via Least Squares (Minimum Norm)

Let $F(Ku, f) = \frac{1}{2} \|Ku - f\|_{L^2(\Sigma)}^2$.

Examples of Selection Operators (continued)

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Let $F(Ku, f) = \frac{1}{2} \|Ku - f\|_{L^2(\Sigma)}^2$. Best approximate solutions are least-squares solutions, satisfying $K^*Ku = K^*f$.

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How to Compute Selection Operators

Let $K : \mathcal{U} \rightarrow \mathcal{V}$ for Banach spaces $\mathcal{U}$ and $\mathcal{V}$.

Selection via Exact Fit (J-minimising solution)

Suppose we choose $F(Ku, f) = \chi_{=0}(f - Ku)$ as earlier.

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for a proper, lower semi-continuous and convex functional $J : \mathcal{U} \rightarrow \mathbb{R}_+ \cup \{+\infty\}$.

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for a proper, lower semi-continuous and convex functional $J : \mathcal{U} \rightarrow \mathbb{R}_+ \cup \{+\infty\}$. In primal-dual form, the corresponding saddle point problem reads

$$\inf_{u \in \mathcal{U}} \sup_{v \in \mathcal{V}} J(u) + \langle v, f - Ku \rangle.$$

How to Compute Selection Operators (continued)

Let $K : \mathcal{U} \rightarrow \mathcal{V}$ for Banach spaces $\mathcal{U}$ and $\mathcal{V}$.

Selection via Exact Fit (J-minimising solution)

In primal-dual form, the corresponding saddle point problem reads

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How do we compute $u^\dagger$ and $v^\dagger$ numerically?

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One option: Primal-Dual Hybrid Gradient / Chambolle-Pock algorithm [18, 13, 6, 3, 4], i.e.

$$\begin{aligned} u^{k+1} &= \text{prox}_{\tau J} (u^k + \tau K^* v^k), \\ v^{k+1} &= v^k - \sigma (K(2u^{k+1} - u^k) - f), \end{aligned}$$

for parameters $\tau, \sigma > 0$ with $\tau\sigma < 1/\|K\|^2$.

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with $\tau\sigma < 1/\|K\|^2$. You will compute J-minimising solutions in your first lab work!

How to Compute Selection Operators (continued)

Example: Let $K = I$ be the identity operator and $J = \text{TV}$ the (isotropic) total variation.



$f = u^\dagger$ (ground truth)

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How to Compute Selection Operators (continued)

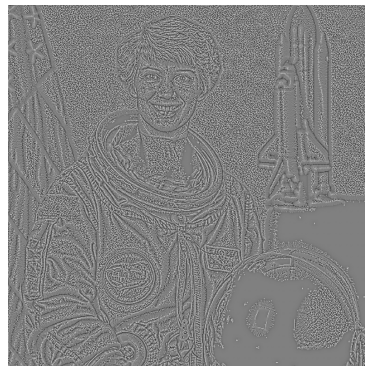
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$v^\dagger$ (source condition element)

How to Compute Selection Operators (continued)

Example: Let $K = \cdot * h$ be a motion blur operator and $J = \text{TV}$ the (isotropic) total variation.



$$f = u^\dagger * h \text{ (blurred image)}$$

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Convergent Regularisation Methods

Convergent Regularisation Method: Definition

Recall: A regularisation method consists of regularisation operators R_α and a parameter choice strategy $\alpha_{\text{choice}}(\delta, f^\delta)$.

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Convergent Regularisation Method [1]

A regularisation method is called **convergent** if for any "exact" data f (for which a set of desired solutions $\mathcal{S}(f) \subset \mathcal{U}$ is well-defined), any sequence of noise levels $\delta_n \rightarrow 0$, data f^{δ_n} satisfying $F(f, f^{\delta_n}) \leq \delta_n$ and parameter choice strategy $\alpha_n = \alpha_{\text{choice}}(\delta_n, f^{\delta_n})$ we have

$$\emptyset \neq \left\{ x \in \mathcal{U}, \left| \limsup_{n \rightarrow \infty} d_{\mathcal{U}}(x, R_{\alpha_n}(f^{\delta_n})) = 0 \right. \right\} \subset \mathcal{S}(f)$$

This means the Kuratowski limit inferior of $R_{\alpha_n}(f^{\delta_n})$ must be non-empty and contained in the set of desired solutions $\mathcal{S}(f)$.

Convergent Regularisation Method: Definition

Recall: A regularisation method consists of regularisation operators R_α and a parameter choice strategy $\alpha_{\text{choice}}(\delta, f^\delta)$. If R_α is single-valued, the previous definition simplifies to:

Convergent Regularisation Method (Single-Valued Operators)

A regularisation method is called **convergent** if for any "exact" data f (for which a set of desired solutions $\mathcal{S}(f) \subset \mathcal{U}$ is well-defined), any sequence of noise levels $\delta_n \rightarrow 0$, data f^{δ_n} satisfying $F(f, f^{\delta_n}) \leq \delta_n$ and parameter choice strategy $\alpha_n = \alpha_{\text{choice}}(\delta_n, f^{\delta_n})$ we have

$$\lim_{n \rightarrow \infty} R_{\alpha_n}(f^{\delta_n}) = u^* \quad \text{where } u^* \in \mathcal{S}(f).$$

This means that the sequence of solutions $R_{\alpha_n}(f^{\delta_n})$ must converge to some solution u^* , and $u^* \in \mathcal{S}(f)$.

Example: Tikhonov Regularisation

Let us revisit Tikhonov regularisation. Recall that the filter functions $g_\alpha(\sigma)$ are defined as

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Key Property for Fixed $\alpha > 0$

For any fixed $\alpha > 0$, the filter function $g_\alpha(\sigma)$ is bounded, i.e.

$$\sup_{\sigma > 0} \left| \frac{\sigma}{\sigma^2 + \alpha} \right| = \frac{1}{2\sqrt{\alpha}} < \infty$$

Thus, R_α is a bounded (and therefore continuous) linear operator for fixed $\alpha > 0$.

Tikhonov as a Regularisation Operator

We recall that Tikhonov regularisation fits the definition of a regularisation operator.

Recall: Regularisation Operator (Single-Valued Case)

An operator $R_\alpha : \mathcal{V} \rightarrow \mathcal{U}$ is a regularisation operator if it is **continuous** on $\mathcal{V}$ for each fixed α .

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Tikhonov is a Regularisation Operator

- For any fixed $\alpha > 0$, the Tikhonov filter function $g_\alpha(\sigma) = \frac{\sigma}{\sigma^2 + \alpha}$ is bounded, as $\sup_{\sigma > 0} |g_\alpha(\sigma)| = \frac{1}{2\sqrt{\alpha}} =: C_\alpha < \infty$.
- This ensures that the operator $R_\alpha f = \sum_{j=1}^{\infty} g_\alpha(\sigma_j) \langle f, v_j \rangle_{\mathcal{V}} u_j$ is a bounded linear operator, because $\|R_\alpha f\|_{\mathcal{U}} \leq C_\alpha \|f\|_{\mathcal{V}}$.
- Bounded linear operators are continuous.
- Hence, for each fixed $\alpha > 0$, R_α is a **regularisation operator**.

Tikhonov as a Convergent Regularisation Method

Let $u^\dagger = K^\dagger f$ be the desired (minimum-norm least-squares) solution for exact data $f \in \mathcal{D}(K^\dagger)$. So $\mathcal{S}(f) = \{K^\dagger f\}$.

Recall: Convergent Regularisation Method (Single-Valued)

$(R_\alpha, \alpha_{\text{choice}})$ is convergent if for noise $\delta_n \rightarrow 0$ and data f^{δ_n} (with $F(f, f^{\delta_n}) \leq \delta_n$), setting $\alpha_n = \alpha_{\text{choice}}(\delta_n, f^{\delta_n})$, we have $\lim_{n \rightarrow \infty} R_{\alpha_n}(f^{\delta_n}) = K^\dagger f$.

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Convergence of Tikhonov Regularisation

For Tikhonov regularisation, the filter $g_\alpha(\sigma_j) = \frac{\sigma_j}{\sigma_j^2 + \alpha}$ satisfies

- $\lim_{\alpha \rightarrow 0} g_\alpha(\sigma_j) = \lim_{\alpha \rightarrow 0} \frac{\sigma_j}{\sigma_j^2 + \alpha} = \frac{1}{\sigma_j}$ for $\sigma_j > 0$. This is a key condition for approximating $K^\dagger$.

Tikhonov as a Convergent Regularisation Method

Convergence of Tikhonov Regularisation (continued)

Consider an a-priori parameter choice $\alpha(\delta)$ that depends on the noise level δ . The error is

$$\left\| R_{\alpha(\delta)} f^\delta - K^\dagger f \right\|_{\mathcal{U}} \leq \left\| R_{\alpha(\delta)} (f^\delta - f) \right\|_{\mathcal{U}} + \left\| R_{\alpha(\delta)} f - K^\dagger f \right\|_{\mathcal{U}} .$$

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- $\left\| R_{\alpha(\delta)} (f^\delta - f) \right\|_{\mathcal{U}} \leq \left\| R_{\alpha(\delta)} \right\|_{\mathcal{L}(\mathcal{V}, \mathcal{U})} \left\| f^\delta - f \right\|_{\mathcal{V}} \leq \frac{1}{2\sqrt{\alpha(\delta)}} \delta.$

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- $\left\| R_{\alpha(\delta)} f - K^\dagger f \right\|_{\mathcal{U}} = \left\| \sum_{j=1}^{\infty} \left(\frac{\sigma_j}{\sigma_j^2 + \alpha(\delta)} - \frac{1}{\sigma_j} \right) \langle f, v_j \rangle_{\mathcal{V}} u_j \right\|_{\mathcal{U}} \rightarrow 0$ as $\alpha(\delta) \rightarrow 0$ if $f \in \mathcal{D}(K^\dagger).$

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Thus, if $\alpha(\delta) \rightarrow 0$ AND $\delta/\sqrt{\alpha(\delta)} \rightarrow 0$ as $\delta \rightarrow 0$, then $R_{\alpha(\delta)} f^\delta \rightarrow K^\dagger f$. Tikhonov regularisation with such an $\alpha(\delta)$ is a **convergent regularisation method**.

Variational Regularisation Methods

Variational Regularisation: A Major Class of Methods

A significant and widely used approach to constructing regularisation operators is through **variational regularisation**.

Variational Regularisation Operator

The (potentially set-valued) operator $R_\alpha : \mathcal{V} \rightrightarrows \mathcal{U}$ defined as

$$R_\alpha(f^\delta) := \arg \min_{u \in \mathcal{U}} \{F(Ku, f^\delta) + J_\alpha(u)\}$$

is said to be a **variational regularisation**. Here $\alpha \in A$ are the regularisation parameter(s).

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We have

- A **data fidelity term**, $F(Ku, f^\delta)$, measuring how well Ku fits the observed data f^δ .
- A (parameterised) **regularisation term** (or penalty term), $J_\alpha(u)$, which incorporates prior knowledge about the desired solution u (e.g., smoothness, sparsity).

Variational Regularisation: Key Theoretical Assumptions

Assumption (Based on [1, Assumption 5.4])

Let $\mathcal{U} = Z^*$ for some normed space Z , and let the weak-star topology on $\mathcal{U}$ be metrisable on bounded sets. Moreover assume

- $K = L^*$ for a bounded linear operator $L : \mathcal{V} \rightarrow Z$.

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- F is proper, non-negative, convex functional in first argument, and continuous in second argument; for every $g \in \mathcal{V}$ there exists $u \in \mathcal{U}$ such that $F(Ku, g) + J_\alpha(u) < \infty$.

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- F is proper, non-negative, convex functional in first argument, and continuous in second argument; for every $g \in \mathcal{V}$ there exists $u \in \mathcal{U}$ such that $F(Ku, g) + J_\alpha(u) < \infty$.
- For each $g \in \mathcal{V}$ and $\alpha \in A$, there exists a constant $c = c(\alpha, b, \|g\|_{\mathcal{V}})$, which depends monotonically non-decreasingly on all its arguments, such that

$$\|u\|_{\mathcal{U}} \leq c \quad \text{if } F(Ku, g) \leq \alpha \text{ and } J_\alpha(u) \leq b.$$

Variational Regularisation: Operator Stability

Under those assumptions we can guarantee

Well-Posedness of $R_\alpha(f^\delta)$

For every $f^\delta \in \mathcal{V}$ and $\alpha \in A$, the set of minimisers $R_\alpha(f^\delta)$ is non-empty. If $F(Ku, f^\delta) + J(u, \alpha)$ is strictly convex, the minimiser is unique.

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Stability of $R_\alpha(f^\delta)$ (Operator Property)

If F is continuous w.r.t. its second variable, and $f^{\delta_n} \rightarrow f^\delta$ in $\mathcal{V}$:

- Any sequence $u_n \in R_\alpha(f^{\delta_n})$ possesses a (weakly, weak-*, or strongly, depending on space and functional properties) convergent subsequence $u_{n_k} \rightarrow u^*$.
- Crucially, this limit point u^* is itself a minimiser for the limit data: $u^* \in R_\alpha(f^\delta)$.

This implies $\emptyset \neq \{u \in \mathcal{U} \mid \limsup_{k \rightarrow \infty} d_{\mathcal{U}}(u, R_\alpha(f^{\delta_k})) = 0\} \subset R_\alpha(f^\delta)$.

Variational Regularisation: Examples

Tikhonov Regularisation (Revisited)

$$\begin{aligned} R_{\alpha}(f^{\delta}) &= \arg \min_{u \in \mathcal{U}} \left\{ \frac{1}{2} \|Ku - f^{\delta}\|_{\mathcal{V}}^2 + \frac{\alpha}{2} \|u\|_{\mathcal{U}}^2 \right\}, \\ &= (K^*K + \alpha I)^{-1} K^* f^{\delta}. \end{aligned}$$

Here $F(Ku, f^{\delta}) = \frac{1}{2} \|Ku - f^{\delta}\|_{\mathcal{V}}^2$ and $J_{\alpha}(u) = \frac{\alpha}{2} \|u\|_{\mathcal{U}}^2$. This fits the variational framework and is a convergent method with appropriate $\alpha_{\text{choice}}(\delta, f^{\delta})$.

Variational Regularisation: Examples

LASSO (Least Absolute Shrinkage and Selection Operator)

$$R_{\alpha}(f^{\delta}) = \arg \min_{u \in \mathbb{R}^n} \left\{ \frac{1}{2} \|Ku - f^{\delta}\|_{\mathbb{R}^m}^2 + \alpha \|u\|_{\ell^1} \right\}$$

Here $J_{\alpha}(u) = \alpha \|u\|_{\ell^1} = \alpha \sum_i^n |u_i|$. Promotes sparse solutions. Its convergence analysis relies on convexity and properties of the ℓ^1 -norm.

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Total Variation (TV) Regularisation (e.g., for Images with sharp edges)

$$R_{\alpha}(f^{\delta}) = \arg \min_u \left\{ \frac{1}{2} \|Ku - f^{\delta}\|^2 + \alpha \text{TV}(u) \right\}$$

where $\text{TV}(u) = \sup_{\varphi \in \{\phi \in C_0^{\infty}(\Omega; \mathbb{R}^n) \mid \|\phi(x)\|_2 \leq 1\}} \int_{\Omega} u(x) \operatorname{div} \varphi(x) \, dx$.

Convergence Analysis: Error Estimates

Error Estimates: Motivation for Deeper Analysis

Once we establish that a regularisation method is *convergent* (e.g., $R_{\alpha_n}(f^{\delta_n}) \rightarrow u^*$ for $u^* \in \mathcal{S}(f)$ as noise $\delta_n \rightarrow 0$ in the single-valued cases), further important questions arise:

- **How good is the approximation?** We need a way to measure the error between the regularised solution $R_\alpha(f^\delta)$ and the desired true solution $u^\dagger$.

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- **How good is the approximation?** We need a way to measure the error between the regularised solution $R_\alpha(f^\delta)$ and the desired true solution $u^\dagger$.
- **How fast does the error decrease** as the noise level δ vanishes? This refers to the *rate of convergence*.

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To address these, we introduce an error measure $D : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}_+ \cup \{+\infty\}$ in the solution space $\mathcal{U}$. This D is not necessarily a norm (e.g., it could be a Bregman distance).

D-Convergence of a Regularisation Method

D-convergent

Let $D : \mathcal{U} \times \mathcal{U} \rightarrow \mathbb{R}_+ \cup \{+\infty\}$ be an error measure. Let $u^\dagger \in \mathcal{S}(f)$ be a desired inverse problem solution corresponding to exact data f . A regularisation method (consisting of operators R_α and parameter choice $\alpha_{\text{choice}}(\delta, f^\delta)$) is called **D-convergent** if for any $u_{\alpha(\delta, f^\delta)}^\delta \in R_{\alpha(\delta, f^\delta)}(f^\delta)$ we observe

$$\lim_{\delta \rightarrow 0} \sup \left\{ D(u_{\alpha(\delta, f^\delta)}^\delta, u^\dagger) \mid f^\delta \in \mathcal{V}, F(f, f^\delta) \leq \delta \right\} = 0.$$

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- $F(f, f^\delta) \leq \delta$ models the noise in the data f^δ .
- This means the maximum error (measured by D) between any obtained regularised solution and the true solution $u^\dagger$ vanishes as the noise level δ goes to zero.

Convergence Rates of a Regularisation Method

To discuss specific rates, we often restrict the "true" solution $u^\dagger$ to a *smoothness class* $\mathcal{M}_\nu \subset \mathcal{U}$, where $\nu > 0$ measures the degree of smoothness or regularity.

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Convergent at Order ν

A regularisation method is called **convergent at order ν** on a set $\mathcal{M}_\nu$ if, for all f such that $Ku^\dagger = f$ for some $u^\dagger \in \mathcal{M}_\nu$, there exists a constant $C_\nu > 0$ such that for all data f^δ satisfying $F(f, f^\delta) \leq \delta$:

$$\sup_{u \in R_{\alpha(\delta, f^\delta)}(f^\delta)} D(u, u^\dagger) \leq C_\nu \delta^\nu$$

Convergence Rates of a Regularisation Method

To discuss specific rates, we often restrict the "true" solution $u^\dagger$ to a *smoothness class* $\mathcal{M}_\nu \subset \mathcal{U}$, where $\nu > 0$ measures the degree of smoothness or regularity.

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- This provides a quantitative estimate on how fast the error decreases as a function of the noise level δ , for solutions $u^\dagger$ possessing sufficient regularity (i.e., $u^\dagger \in \mathcal{M}_\nu$).

Tikhonov: Convergence Rate

We use the definition of "Convergent at Order ν " with error measure $D(u, v) = \|u - v\|_{\mathcal{U}}$.

Smoothness Assumption (Source Condition)

Assume the true solution $u^\dagger = K^\dagger f$ possesses a certain "smoothness". A common assumption is $u^\dagger \in \mathcal{R}((K^*K)^\mu)$ for some $\mu > 0$, i.e.,
 $u^\dagger = (K^*K)^\mu w = \sum_{j=1}^{\infty} \sigma_j^{2\mu} \langle w, u_j \rangle_{\mathcal{U}} u_j$, for some $w \in \mathcal{U}$.

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Under this smoothness assumption ($\mu > 0$), for $u_\alpha^\delta = R_\alpha f^\delta$ (Tikhonov), the error can be bounded. The two main error components behave as:

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Under this smoothness assumption ($\mu > 0$), for $u_{\alpha}^{\delta} = R_{\alpha} f^{\delta}$ (Tikhonov), the error can be bounded. The two main error components behave as:

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Balancing these terms by choosing $\alpha(\delta) \sim \delta^{2/(2\mu+1)}$ gives

$$\|R_{\alpha(\delta)}f^{\delta} - u^{\dagger}\|_{\mathcal{U}} \leq C\delta^{\frac{2\mu}{2\mu+1}}.$$

Hence, $R_{\alpha(\delta)}$ is **convergent at order $\nu = 2\mu/(2\mu + 1)$** for solutions $u^{\dagger} \in \mathcal{R}((K^{*}K)^{\mu})$, $\mu \in (0, 1]$. For $\mu = 1$, the rate is $\mathcal{O}(\delta^{2/3})$. If $u^{\dagger} \in \mathcal{R}(K^{*})$ (i.e. $\mu = 1/2$), the rate is $\mathcal{O}(\sqrt{\delta})$.

Beyond Norms: Bregman Distances for Error Measurement

The error measure $D(u, v)$ in convergence analysis need not be a norm. **Bregman distances** [2, 11] offer powerful alternatives for analysing variational regularisations.

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Bregman Distance

Let $J : \mathcal{U} \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper, convex and lower semi-continuous functional. For $u_1, u_2 \in \mathcal{U}$ and $p_2 \in \partial J(u_2)$ (a subgradient of J at u_2), the Bregman distance is defined as

$$D_J^{p_2}(u_1, u_2) = J(u_1) - J(u_2) - \langle p_2, u_1 - u_2 \rangle.$$

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Bregman Distance

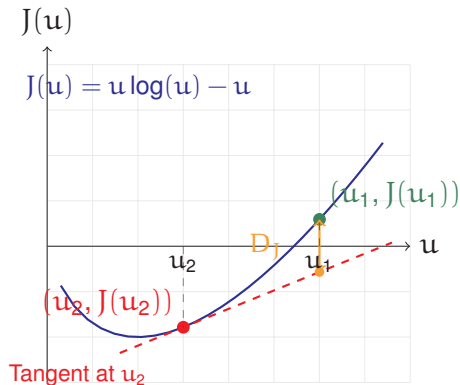
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- Non-negative: $D_J^{p_2}(u_1, u_2) \geq 0$
- Generally non-symmetric: $D_J^{p_2}(u_1, u_2) \neq D_J^{p_1}(u_2, u_1)$
- Symmetric special case: If $J(u) = \frac{1}{2} \|u\|^2$, then $D_J^{u_2}(u_1, u_2) = \frac{1}{2} \|u_1 - u_2\|^2$

Visualising Bregman Distance: $J(u) = u \log(u) - u$



Interpretation

- **Function:** Convex for $u > 0$
- **Derivative:** $J'(u) = \log(u)$
- **Subgradient:** $p_2 = \log(u_2)$ at point u_2

Bregman Distance

$$D_J^{\log(u_2)}(u_1, u_2) = u_1 \log\left(\frac{u_1}{u_2}\right) + u_2 - u_1$$

This is the **Kullback-Leibler divergence** between u_1 and u_2 .

Symmetrised Bregman Distances

To obtain a symmetric measure, one can define the **symmetrised Bregman distance**:

Symmetrised Bregman Distance

Given $p \in \partial J(u)$ and $q \in \partial J(v)$ we define

$$\begin{aligned} D_J^{\text{symm}}(u, v) &= D_J^p(v, u) + D_J^q(u, v) \\ &= J(v) - J(u) - \langle p, v - u \rangle + J(u) - J(v) - \langle q, u - v \rangle \\ &= \langle q - p, v - u \rangle \geq 0. \end{aligned}$$

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Relevance

Symmetrised Bregman distances naturally appear in error estimates for variational regularisation methods, particularly when source conditions are involved.

Recap: Error Estimates & Bregman Distances

We are interested in error estimates for variational regularisation with scalar parameter, i.e.

$$R_{\alpha}(f^{\delta}) = u_{\alpha}^{\delta} \in \arg \min_{u \in \mathcal{U}} \left\{ \frac{1}{2} \|Ku - f^{\delta}\|_{\mathcal{V}}^2 + \alpha J(u) \right\}$$

Let $u^{\dagger}$ be the true solution with exact data $f = Ku^{\dagger}$.

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Key Ingredients

- **Optimality Condition for u_α^δ :** there exists $p_\alpha \in \partial J(u_\alpha^\delta)$ such that

$$K^*(Ku_\alpha^\delta - f^\delta) + \alpha p_\alpha = 0.$$

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- **Optimality Condition for u_α^δ :** there exists $p_\alpha \in \partial J(u_\alpha^\delta)$ such that

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- **Source Condition for $u^\dagger$** (assumed to hold for $u^\dagger$): there exists $v \in \mathcal{V}$ such that

$$K^*v \in \partial J(u^\dagger).$$

Deriving Error Estimates: Step 1 (Main Equation)

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- 4 Recognising that $\langle p_\alpha - K^*v, u_\alpha^\delta - u^\dagger \rangle_u = D_J^{\text{symm}}(u_\alpha^\delta, u^\dagger)$, where the specific subgradients $p_\alpha \in \partial J(u_\alpha^\delta)$ and $K^*v \in \partial J(u^\dagger)$ are used

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- 5 Hence, with $f = Ku^\dagger$, the previous equation becomes

$$\langle Ku_\alpha^\delta - f^\delta, Ku_\alpha^\delta - f \rangle_v + \alpha D_J^{\text{symm}}(u_\alpha^\delta, u^\dagger) = \alpha \langle v, f - Ku_\alpha^\delta \rangle_v$$

Deriving Error Estimates: Step 2 (Using Identities)

From the previous slide we have the equation

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We use two standard algebraic identities for inner products:

- 1 $\langle a - b, a - c \rangle_{\mathcal{V}} = \frac{1}{2} \|a - c\|_{\mathcal{V}}^2 + \frac{1}{2} \|a - b\|_{\mathcal{V}}^2 - \frac{1}{2} \|c - b\|_{\mathcal{V}}^2$. Applying this to the first term with $a = Ku_{\alpha}^{\delta}$, $b = f^{\delta}$, $c = f$ yields

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- 2** $\langle \mathbf{x}, \mathbf{y} \rangle_{\mathcal{V}} = \frac{1}{2} \|\mathbf{x}\|_{\mathcal{V}}^2 + \frac{1}{2} \|\mathbf{y}\|_{\mathcal{V}}^2 - \frac{1}{2} \|\mathbf{x} - \mathbf{y}\|_{\mathcal{V}}^2$. Applying this to the right-hand-side yields

$$\alpha \langle \mathbf{v}, \mathbf{f} - \mathbf{K}\mathbf{u}_\alpha^\delta \rangle_{\mathcal{V}} = \frac{\alpha^2}{2} \|\mathbf{v}\|_{\mathcal{V}}^2 + \frac{1}{2} \|\mathbf{K}\mathbf{u}_\alpha^\delta - \mathbf{f}\|_{\mathcal{V}}^2 - \frac{1}{2} \|\mathbf{K}\mathbf{u}_\alpha^\delta - \mathbf{f} + \alpha \mathbf{v}\|_{\mathcal{V}}^2 .$$

Deriving Error Estimates: Step 3 (Combining)

Substituting the identities into the main equation from Step 1 leads to

$$\begin{aligned} & \left(\frac{1}{2} \|Ku_\alpha^\delta - f\|_V^2 + \frac{1}{2} \|Ku_\alpha^\delta - f^\delta\|_V^2 - \frac{1}{2} \|f - f^\delta\|_V^2 \right) + \alpha D_J^{\text{symm}}(u_\alpha^\delta, u^\dagger) \\ &= \left(\frac{\alpha^2}{2} \|v\|_V^2 + \frac{1}{2} \|Ku_\alpha^\delta - f\|_V^2 - \frac{1}{2} \|Ku_\alpha^\delta - f + \alpha v\|_V^2 \right) \end{aligned}$$

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The term $\frac{1}{2} \|Ku_\alpha^\delta - f\|_V^2$ cancels on both sides. Rearranging leaves us with

$$\frac{1}{2} \|Ku_\alpha^\delta - f + \alpha v\|_V^2 + \frac{1}{2} \|Ku_\alpha^\delta - f^\delta\|_V^2 + \alpha D_J^{\text{symm}}(u_\alpha^\delta, u^\dagger) = \frac{1}{2} \|f - f^\delta\|_V^2 + \frac{\alpha^2}{2} \|v\|_V^2$$

Deriving Error Estimates: Step 4 (Bounding)

The Error Bound

Since the first two terms on the left-hand-side are non-negative, we observe

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With the noise estimate $\|\mathbf{f} - \mathbf{f}^\delta\|_{\mathcal{V}} \leq \delta$ we further obtain

$$D_J^{\text{symm}}(\mathbf{u}_\alpha^\delta, \mathbf{u}^\dagger) \leq \frac{\delta^2}{2\alpha} + \frac{\alpha}{2} \|\mathbf{v}\|_{\mathcal{V}}^2$$

This is a common form of error estimate for variational regularisation (for quadratic fidelity terms).

D-Convergence and Rates from Bregman Estimate

We have derived $D_J^{\text{symm}}(u_\alpha^\delta, u^\dagger) \leq \frac{\delta^2}{2\alpha} + \frac{\alpha}{2} \|v\|_{\mathcal{V}}^2$. Using $D(\cdot, \cdot) = D_J^{\text{symm}}(\cdot, \cdot)$ as our error measure, we can achieve D-convergence.

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Achieving D-Convergence and Rates

- **D-Convergence:** If the parameter choice strategy ensures $\alpha(\delta) \rightarrow 0$ and $\delta^2/\alpha(\delta) \rightarrow 0$ as $\delta \rightarrow 0$, then the right-hand-side converges to zero. This implies $D_J^{\text{symm}}(u_{\alpha(\delta)}^\delta, u^\dagger) \rightarrow 0$, making the method D-convergent.

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- **Convergence Rate (Example: Order $\nu = 1$):** To optimise the bound, choose $\alpha(\delta) = \frac{\delta}{\|\mathbf{v}\|_\mathcal{V}}$, which yields

$$D_J^{\text{symm}}(\mathbf{u}_{\alpha(\delta)}^\delta, \mathbf{u}^\dagger) \leq \|\mathbf{v}\|_\mathcal{V} \delta$$

This shows convergence at order $\nu = 1$ with rate constant $C_1 = \|\mathbf{v}\|_\mathcal{V}$.

Iterative Regularisation Methods

Iterative Regularisation: Motivation

We've explored variational methods of the form:

$$\mathbf{u}_\alpha^\delta \in \mathbf{R}_\alpha(\mathbf{f}^\delta) = \arg \min_{\mathbf{u} \in \mathcal{U}} \{F(K\mathbf{u}, \mathbf{f}^\delta) + \alpha J(\mathbf{u})\}$$

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- **Question:** Can an iterative algorithm itself, when applied to f^{δ} and stopped early, act as a regularisation method?

Iterative Regularisation: Motivation

We've explored variational methods of the form:

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Example: PDHG for J-Minimising Solutions with Noisy Data

We'll examine the PDHG algorithm for finding a J-minimising solution to $Ku = f$, even when we only have noisy data f^δ (cf. [12]).

PDHG for J-Minimising Solutions with Noisy Data

Recall the problem of finding a J-minimising solution:

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Algorithm with Noisy Data f^δ

The PDHG algorithm for this problem, using noisy data f^δ where $\|f - f^\delta\|_{\mathcal{V}} \leq \delta$ reads

$$\begin{aligned} u^{k+1} &= \text{prox}_{\tau J} (u^k + \tau K^* v^k) \\ v^{k+1} &= v^k - \sigma (K(2u^{k+1} - u^k) - f^\delta) \end{aligned}$$

with $\tau, \sigma > 0$ such that $\tau\sigma < 1/\|K\|^2$. We assume $u^0 = 0, v^0 = 0$.

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with $\tau, \sigma > 0$ such that $\tau\sigma < 1/\|K\|^2$. We assume $u^0 = 0, v^0 = 0$.

Goal: Show that producing $\bar{u}^k$ (e.g., Cesàro mean of u^j for $j = 1, \dots, k$) with $k = k^*(\delta)$ (an early stopping rule) makes this a convergent regularisation method.

Convergence Analysis: Key Ingredients

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- Define a symmetric operator $M : \mathcal{U} \times \mathcal{V} \rightarrow \mathcal{U} \times \mathcal{V}$ and its associated M -norm

$$M := \begin{pmatrix} \frac{1}{\tau} I & -K^* \\ -K & \frac{1}{\sigma} I \end{pmatrix}, \quad \|w\|_M^2 := \langle Mw, w \rangle.$$

Note: for $\tau\sigma < 1/\|K\|^2$, M is (symmetric) positive definite.

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- Then, the updates of the PDHG algorithm satisfy the equivalent system of optimality conditions

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- The J-minimising solution $u^\dagger$ and corresponding source condition element $v^\dagger$ satisfy

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} \in \begin{pmatrix} \partial J(u^\dagger) - K^* v^\dagger \\ Ku^\dagger - f \end{pmatrix}.$$

Convergence Analysis: Key Ingredients

Subtracting one condition from the other yields

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Taking the dual product with $w^k - w^\dagger$, we obtain

$$0 = D_J^{\text{symm}}(u^k, u^\dagger) + \langle f - f^\delta, v^k - v^\dagger \rangle + \langle M(w^k - w^{k-1}), w^k - w^\dagger \rangle,$$

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respectively

$$0 \leq D_J^{\text{symm}}(u^k, u^\dagger) = \langle M(w^{k-1} - w^k), w^k - w^\dagger \rangle + \langle f^\delta - f, v^k - v^\dagger \rangle.$$

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Hence, if we define $\tilde{w} := M^{-1} \begin{pmatrix} 0 \\ f^\delta \end{pmatrix}$, we estimate

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Convergence Analysis: Key Ingredients

This means we have verified

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M-Norm Estimate

For a given positive definite operator M and corresponding inner product and norm, the inequality $\langle M(w^k - w^{k-1}), w^k - w^\dagger \rangle \leq \frac{\delta}{\sqrt{\|M\|}} \|w^k - w^\dagger\|_M$ implies

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Method of Induction

$$\|w^k - w^\dagger\|_M \leq \|w^\dagger\|_M + k \delta / \sqrt{\|M\|}.$$

Convergence Analysis: Summing the Estimates

Together with the three-point identity

$$\left\langle M(w^{k-1} - w^k), w^k - w^\dagger \right\rangle = \frac{1}{2} \|w^{k-1} - w^\dagger\|_M^2 - \frac{1}{2} \|w^k - w^\dagger\|_M^2 - \frac{1}{2} \|w^{k-1} - w^k\|_M^2,$$

we estimate with previous result

$$\begin{aligned} & D_J^{\text{symm}}(u^k, u^\dagger) + \frac{1}{2} \|w^k - w^{k-1}\|_M^2 \\ & \leq \frac{1}{2} \|w^{k-1} - w^\dagger\|_M^2 - \frac{1}{2} \|w^k - w^\dagger\|_M^2 + \frac{\delta \|w^\dagger\|_M}{\sqrt{\|M\|}} + \frac{k\delta^2}{\|M\|}. \end{aligned}$$

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Summing up from 1 to k then yields

$$\sum_{j=1}^k \left[D_J^{\text{symm}}(u^j, u^\dagger) + \frac{1}{2} \|w^j - w^{j-1}\|_M^2 \right] \leq \frac{1}{2} \|w^\dagger\|_M^2 + \frac{k\delta \|w^\dagger\|_M}{\sqrt{\|M\|}} + \frac{k(k+1)\delta^2}{2\|M\|}.$$

Convergence Rate via Early Stopping

Let $\bar{u}^k = \frac{1}{k} \sum_{j=1}^k u^j$ be the Cesaro mean of the primal iterates. Since J is convex, $D_J^{\text{symm}}(\cdot, u^\dagger)$ is also convex in its first argument. By Jensen's inequality:

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and dividing by k yields

$$D_J^{\text{symm}}(\bar{u}^k, u^\dagger) \leq \frac{\|w^\dagger\|_M^2}{2k} + \frac{\delta \|w^\dagger\|_M}{\sqrt{\|M\|}} + \frac{(k+1)\delta^2}{2\|M\|}.$$

Order of Convergence for PDHG

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Choose number of iterations $k = k^*(\delta)$ such that $k^*(\delta) \sim 1/\delta$, e.g. $k^*(\delta) = \left\lceil \frac{\sqrt{\|M\|} \|w^\dagger\|_M}{\delta} \right\rceil$.

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Convergence Result

With an appropriate stopping rule $k^*(\delta) \sim 1/\delta$, the Condat-Vu algorithm is D-convergent with $\nu = 1$ with respect to the symmetrised Bregman distance, i.e.

$$D_J^{\text{symm}}(\bar{u}^{k^*(\delta)}, u^\dagger) = \mathcal{O}(\delta).$$

This means the iterative scheme for computing the selection operator, with f^δ and an early stopping rule $k^*(\delta)$, is a convergent regularisation method.

Iterative Regularisation: Summary

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- This constitutes a convergent regularisation method with rate $\mathcal{O}(\delta)$ for $D_J^{\text{symm}}(\bar{u}^{k^*(\delta)}, u^\dagger)$.
- This links iterative optimisation directly to regularisation theory. Many other iterative methods (Landweber, CG, etc.) also exhibit regularising properties.

Data-Driven Regularisation: Spectral Methods

Data-Driven Spectral Regularisation: Motivation

We've seen spectral regularisation methods $R_\alpha f = \sum_{j=1}^{\infty} g_\alpha(\sigma_j) \langle f, v_j \rangle_v u_j$, where the filter function $g_\alpha(\sigma_j)$ is pre-defined (e.g., Tikhonov, TSVD).

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A New Perspective: Learning the Filter

What if we could *learn* an optimal filter function directly from data?

- Suppose we have a training dataset of true solutions u and corresponding noisy measurements $f^\delta = Ku + \text{noise}$.
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This leads to **data-driven spectral regularisation**, where the filter itself is shaped by the statistical properties of the signal and noise.

Data-Driven Spectral Regularisation: Problem Setup

Let $K : \mathcal{U} \rightarrow \mathcal{V}$ be a compact linear operator between Hilbert spaces with singular system $\{(\sigma_j, u_j, v_j)\}$. We consider measurements $f^\delta = Ku + v$, where u is the true signal and v is additive noise.

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Objective

We seek to find filter coefficients $g_j := g(\sigma_j)$ for the spectral regularisation operator

$$R(f^\delta; g) = \sum_{j=1}^{\infty} g_j \langle f^\delta, v_j \rangle_{\mathcal{V}} u_j$$

by minimising the expected squared error

$$\min_{\{g_j\}} \mathbb{E}_{u,v} \left[\|u - R(f^\delta; g)\|_{\mathcal{U}}^2 \right]$$

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Assumptions for Derivation

- The noise $\mathbf{v}$ has zero mean, i.e. $\mathbb{E}_{\mathbf{v}}[\mathbf{v}] = 0$.
- The noise $\mathbf{v}$ is statistically independent of the signal $\mathbf{u}$.
- The noise components $\langle \mathbf{v}, \mathbf{v}_j \rangle$ are uncorrelated with signal components $\langle \mathbf{u}, \mathbf{u}_k \rangle$.
Specifically, $\mathbb{E}_{\mathbf{u}, \mathbf{v}} [\langle \mathbf{u}, \mathbf{u}_j \rangle \langle \mathbf{v}, \mathbf{v}_j \rangle] = \mathbb{E}_{\mathbf{u}} [\langle \mathbf{u}, \mathbf{u}_j \rangle] \mathbb{E}_{\mathbf{v}} [\langle \mathbf{v}, \mathbf{v}_j \rangle] = 0$ if $\mathbb{E}_{\mathbf{v}} [\langle \mathbf{v}, \mathbf{v}_j \rangle] = 0$.

Data-Driven Spectral Regularisation: Optimal Filter

Under the stated assumptions, the expected squared error can be decomposed:

$$\mathbb{E}_{\mathbf{u}, \mathbf{v}} \left[\|\mathbf{u} - \mathbf{R}(\mathbf{f}^\delta; \mathbf{g})\|_{\mathcal{U}}^2 \right] = \mathbb{E}_{\mathbf{u}} \left[\|\mathbf{u}_0\|_{\mathcal{U}}^2 \right] + \sum_{j=1}^{\infty} \left((1 - \sigma_j g_j)^2 \mathbb{E}_{\mathbf{u}} \left[\langle \mathbf{u}, \mathbf{u}_j \rangle_{\mathcal{U}}^2 \right] + g_j^2 \mathbb{E}_{\mathbf{v}} \left[\langle \mathbf{v}, \mathbf{v}_j \rangle_{\mathcal{V}}^2 \right] \right)$$

where $\mathbf{u}_0$ is the component of $\mathbf{u}$ in $\mathcal{N}(\mathbf{K})$.

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$$\mathbb{E}_{\mathbf{u}, \mathbf{v}} \left[\|\mathbf{u} - \mathbf{R}(\mathbf{f}^\delta; \mathbf{g})\|_{\mathbf{u}}^2 \right] = \mathbb{E}_{\mathbf{u}} \left[\|\mathbf{u}_0\|_{\mathbf{u}}^2 \right] + \sum_{j=1}^{\infty} \left((1 - \sigma_j g_j)^2 \mathbb{E}_{\mathbf{u}} \left[\langle \mathbf{u}, \mathbf{u}_j \rangle_{\mathbf{u}}^2 \right] + g_j^2 \mathbb{E}_{\mathbf{v}} \left[\langle \mathbf{v}, \mathbf{v}_j \rangle_{\mathbf{v}}^2 \right] \right)$$

where $\mathbf{u}_0$ is the component of $\mathbf{u}$ in $\mathcal{N}(K)$. Let us define

- Signal power per component: $\Pi_j := \mathbb{E}_{\mathbf{u}} \left[\langle \mathbf{u}, \mathbf{u}_j \rangle_{\mathbf{u}}^2 \right]$
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$\sum_j \left((1 - \sigma_j g_j)^2 \Pi_j + g_j^2 \Delta_j \right)$. Minimising with respect to each g_j (by setting $\partial/\partial g_j = 0$) yields the optimal filter coefficients

$$\bar{g}_j = \frac{\sigma_j \Pi_j}{\sigma_j^2 \Pi_j + \Delta_j}$$

Data-Driven Spectral Regularisation: Filter Properties

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If $\Pi_j > 0$, we can rewrite this as

$$\bar{g}_j = \frac{\sigma_j}{\sigma_j^2 + (\Delta_j / \Pi_j)}$$

This is identical in form to the Tikhonov filter $g_\alpha(\sigma_j) = \frac{\sigma_j}{\sigma_j^2 + \alpha}$, but with an *adaptive, data-driven regularisation parameter* $\alpha_j = \Delta_j / \Pi_j$ for each singular component j .

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Learned Spectral Method as a Regularisation Operator

Let statistics Π_j and $\Delta_j(\delta)$ (for given overall noise level δ) be fixed. This defines filter coefficients $\bar{g}_j(\delta)$. The operator is $R_{\bar{g}(\delta)}(f) = \sum_j \bar{g}_j(\delta) \langle f, v_j \rangle u_j$.

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Assume there exist $c > 0$ and $j_0 \in \mathbb{N}$ such that for all $j \geq j_0$ and for a given $\delta > 0$ we have

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Thus, $R_{\bar{g}(\delta)}$ is a bounded linear operator for fixed $\delta > 0$, and therefore a **regularisation operator**.

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$$\lim_{\delta \rightarrow 0} \mathbb{E}_{\mathbf{v}} \left[\|\mathbf{u} - \mathbf{R}(f^\delta; \bar{\mathbf{g}}(\delta))\|_{\mathbf{u}}^2 \right] = 0.$$

Furthermore, the expected error over the data distribution also converges, i.e.

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This means the method is convergent in the sense that, on average, the reconstructions approach the true solution as the noise level diminishes.

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For classical spectral methods like Tikhonov or TSVD, convergence rates of the form

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- Deriving explicit $\mathcal{O}(\delta^\nu)$ rates for the deterministic error $\|R(f^\delta; \bar{g}(\delta)) - u^\dagger\|_{\mathcal{U}}$ for the data-driven filter $\bar{g}_j(\delta) = \frac{\sigma_j \Pi_j}{\sigma_j^2 \Pi_j + \Delta_j(\delta)}$ is more involved.
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- The rate depends on:
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- The convergence proof for the MSE (previous slide) demonstrates convergence to zero, but a specific power of δ for the MSE is not immediately extracted without further assumptions on Π_j and $\Delta_j(\delta)$.

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- **Analysis Goals:** Prove stability and convergence, and establish error rates (e.g., $\mathcal{O}(\delta^\nu)$ using norms or Bregman distances).

Outlook and Open Questions

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There are quite a few open problems in the context of deterministic regularisation theory, e.g.

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but we have no convergence rates for their iterative counterparts, i.e.

$$u^{k+1} \in \arg \min_{u \in \mathcal{U}} \left\{ F(Ku, f^\delta) + D_J^{p^k}(u, u^k) \right\} ,$$

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- No convergence rates for coordinate descent-, Kaczmarz or incremental methods, which probably would have closest resemblance to neural network architectures.

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- And many more open questions (discussion?)

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Questions?

Thank You!